

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

21st November 2024

SA7 - Investment and Finance

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q. 1) Pension fund industry has seen Contributory Pension Schemes (CPSs) replacing majority of the erstwhile Defined Benefit Schemes (DBSs) owing to

- increasing fiscal burden of DBSs on governments
- increasing insolvencies of other employers; key reason being deficits in DBSs

People are concerned with CPSs' uncertainty of returns / benefits when compared to DBSs. To promote participation from more people in the CPSs, Pension regulators want to allow some kind of investment guarantee in CPSs. Investment guarantee will be offered through new CPSs by Pension Fund Management Companies (PFMCs) with approval from Pension regulator after assessing the risks and risk mitigation measures. With introduction of investment guarantees, obligations of PFMCs increases, hence Pension regulators now need to enhance monitoring mechanisms. PFMCs are required to regularly submit reports on the guaranteed CPSs.

- i. Describe various types of investment guarantees that could be offered in CPSs. (4)
 - ii. Describe key issues in a guaranteed CPS where ALM (Asset Liability Management) analysis is required. (13)
 - iii. Describe common risks faced by any PFMC. (6)
 - iv. Discuss plausible risks faced by various stakeholders in guaranteed CPSs. (7)
 - v. Suggest ways to control the risks identified in part (iv). (5)
 - vi. List possible indicative reports / statements to be prepared by PFMCs at different frequencies required by Pension Regulations. (15)
- [50]**

Q. 2) You joined a start-up DEF fund house three years ago as assistant fund manager. Your two tactical strategies: Covered Call (CC) and Protective Put (PP) using exchange traded derivatives have played a key role in growth of business. In CC, an investor can increase the income of their fund by writing call options on the assets held. In PP, an investor buys put options on assets held to protect against fall in their market value.

Recently DEF has acquired a thirty year old PQR fund house. You have now become a Deputy CIO (Chief Investment Officer) of the combined entity DEFPQR. The core strategy of both DEF and PQR fund houses was to "Buy and Hold" well researched stocks (equities) for their HNW (High Net Worth) clients. Variable (performance related) pay was given to fund managers who beat their respective sector / factor index set as benchmarks.

Fund House	Average Age of Fund Managers in years	Overall Salary cost split	
		Fixed	Variable
DEF	25	80%	20%
PQR	55	60%	40%

CIO asked you to conduct a training session for fund managers of both DEF and PQR. You will give a presentation about planning and implementing a tactical strategy CC / PP depending upon the research team's short term view on stocks held.

- i. Describe the key points you would include in your presentation for the training session. (7)

After the presentation, DEF fund managers agreed to use the tactical strategies but most of PQR fund managers were reluctant to use covered call. Listening to their concerns you suggested a way (without using derivatives) to enhance portfolio returns.

ii. Discuss possible reasons: why DEF fund managers agreed and PQR fund managers were reluctant to use covered call. (4)

iii. Explain one way (without using derivatives) to enhance portfolio returns. (2)

Chief Marketing Officer has approached you and CIO to discuss rising popularity of the following:

- Sustainable investing with special focus on EV (Electric Vehicles) stocks;
- Smart beta investing; and
- Some exchanges offering derivatives on crypto currencies to reduce tax burden of dealing directly in crypto currencies

iv. Discuss the prospects of EV industry in India by considering the following: (12)

- Factors affecting recent high sales growth rate
- Challenges for future growth rate
- Possible solutions to tackle the challenges

v. Explain smart beta investing. (4)

vi. Discuss a method to create and maintain benchmark index for smart beta investment funds. To illustrate the methodology, you may consider investible universe of stocks from BSE500 index and an example of smart beta fund as Alpha 30 fund. (6)

vii. Describe a merit and the demerits of crypto currency investing in India. (7)

viii. Discuss the usefulness of some exchanges claiming to reduce tax burden by offering derivatives on crypto currencies. (3)

Your research team is analysing XYZ exchange traded fund benchmarked to Nifty 50. XYZ's weekly natural log returns for last five financial years are summarized below:

Percentile	Log Returns		Percentile	Log Returns
0.0	-13.5%		75.0	1.7%
1.0	-6.5%		95.0	3.6%
2.5	-4.7%		97.5	4.5%
5.0	-3.5%		99.0	6.4%
25.0	-0.8%		100.0	10.6%
Mean	0.31%		Standard Deviation	2.43%

ix. For INR 10 lakhs invested in XYZ,

- give two estimates of 95% one week VaR (Value-at-Risk); and
- give an estimate of 95% one year VaR

(5)
[50]
